A Spatially Adaptive Finite State Projection for the Reaction-Diffusion Master Equation

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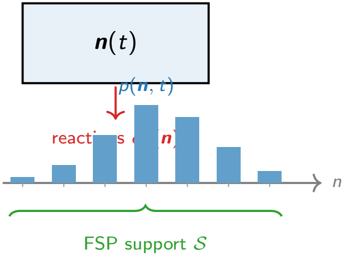
The Chemical Master Equation (CME)

A **single well-mixed compartment** holding molecules of S species. State: molecule count vector $\boldsymbol{n} \in \mathbb{Z}_{\geq 0}^{S}$.

Each reaction r fires with rate $\alpha_r(\boldsymbol{n})$ and changes the state by stoichiometry $\boldsymbol{\nu}_r$. The probability distribution $p(\boldsymbol{n}, t)$ evolves as

$$\frac{dp}{dt} = \underbrace{Ap}_{\text{CME}},$$

where $A_{ij} = \alpha_r(\boldsymbol{n}_j)$ if $\boldsymbol{n}_i = \boldsymbol{n}_j + \boldsymbol{\nu}_r$, and the diagonal ensures column sums are zero.



Finite State Projection (FSP)

Truncate to a finite support $\mathcal{S}$; expand it adaptively as probability mass spreads. State space $|\mathcal{S}|$ grows only as needed — not exponentially in time.

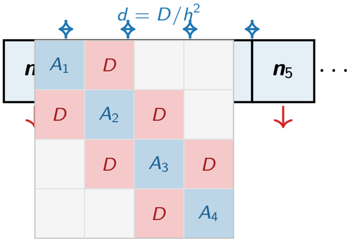
The RDME: K coupled CMEs

Divide the spatial domain into K voxels of width h. Each voxel k has its own count vector $\boldsymbol{n}_k$ and **its own CME**.

Voxels are coupled by **diffusion**: one molecule hops $k \leftrightarrow k\pm1$ at rate $d = D/h^2$ per molecule.

The joint distribution $p(\boldsymbol{n}_1, \dots, \boldsymbol{n}_K, t)$ obeys

$$\frac{dp}{dt} = \underbrace{(A_{\mathrm{rxn}} + A_{\mathrm{diff}})}_{A_{\mathrm{RDME}}}\, p.$$



Exponential cost

Full state space: $n_{\max}^{SK}$.
Even $K{=}8$, $S{=}1$, $n_{\max}{=}50$: intractable.

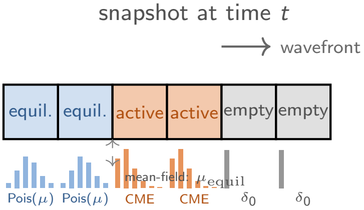
A_{RDME} is **block-tridiagonal**: A_k on diagonal (CME), $D_{k,k+1}$ off-diagonal (diffusion shifts one

Starting from a **localised initial condition** (all molecules in one region), probability mass spreads *diffusively*.

At any time t, the support of p is concentrated near the **diffusion wavefront**: voxels far from the origin are essentially empty.

Three regimes at any instant:

- **Empty**: $p(\boldsymbol{n}_k) \approx \delta_{\boldsymbol{n}_k,\mathbf{0}}$. No tracking needed.
- **Active**: wavefront voxels. Nontrivial distribution — track with the full CME.
- **Equilibrated**: behind the wavefront, distribution has converged to a known form. Store as a single parameter.



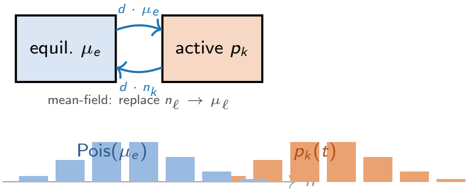
Exact RDME: gain rate into voxel k from neighbor ℓ is $d \cdot n_\ell$, which depends on the *random* count n_ℓ.

Mean-field approximation: replace the random n_ℓ by its mean $\mu_\ell = \mathbb{E}[n_\ell]$, decoupling the voxel CMEs:

$$\frac{dp_k}{dt} = Q_k(\mu_{\mathrm{nbr}})\, p_k,$$

where the effective generator has

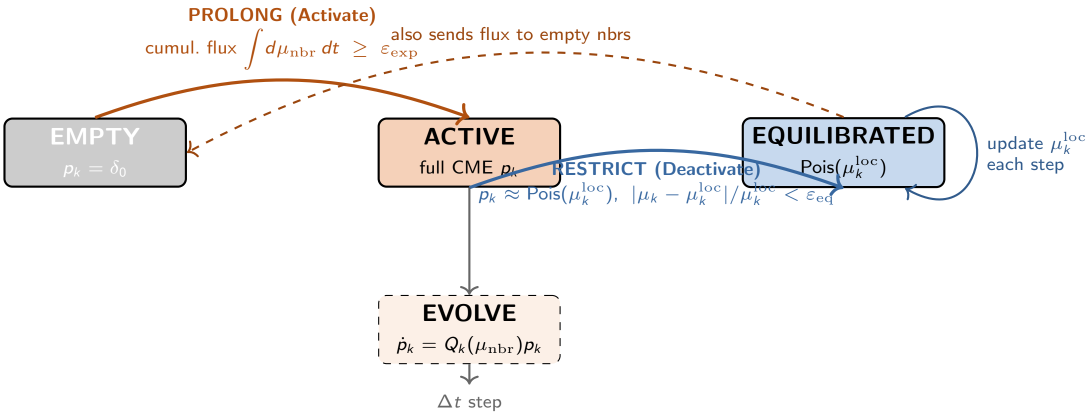
$$k_b^{\mathrm{eff}} = k_b + d \sum_{\ell \sim k} \mu_\ell, \qquad k_d^{\mathrm{eff}} = k_d + d\,|\mathrm{nbrs}|.$$



both converge to Pois(μ^*) = Pois(k_b / k_d)

When is mean-field exact?

For **linear (birth-death) kinetics**, the SS is a product of independent Poissons: $p^* = \prod_k \mathrm{Pois}(\mu^*)$. Detailed balance holds under symmetric diffusion, so



Why is μ_k^{loc} updated each step? The equilibrated voxel's local SS depends on its neighbours: $\mu_k^{\text{loc}}(t) = (k_b + d\sum_\ell \mu_\ell)/(k_d + d|\text{nbrs}|)$. As active neighbours equilibrate, $\mu_k^{\text{loc}} \to k_b/k_d$ (global SS).

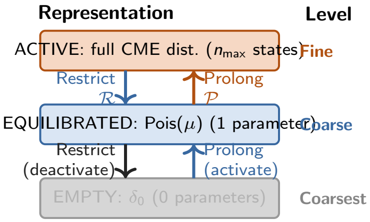
Connection to multigrid: an adaptive spatial V-cycle

Classical geometric multigrid (for linear PDEs):

1. *Pre-smooth*: relax on fine grid.
2. *Restrict* $\mathcal{R}$: fine $\to$ coarse.
3. *Coarse solve*: exact solve on coarse grid.
4. *Prolong* $\mathcal{P}$: coarse $\to$ fine.
5. *Post-smooth*: relax on fine grid.

Spatial adaptive FSP (this work):

1. *Evolve*: CME step on active (fine) voxels.
2. *Restrict*: active $\to$ equilibrated (coarse: Pois(μ)).
3. *Coarse update*: evolve μ_k^{loc} algebraically.
4. *Prolong*: flux threshold $\to$ activate (coarse $\to$ fine).



Three levels of representation with adaptive transitions. Akin to *algebraic* multigrid: the coarsening is driven by the solution (equilibration), not the geometry.

Key difference: adaptive scheduling

In classical MG the cycle is *prescribed* (fixed iteration).

Algorithm: one time step

1. **Compute effective means.**
   Active: $\mu_k = \langle p_k \rangle$. Equilibrated: $\mu_k = \mu_k^{\mathrm{loc}}$ (stored).
2. **Evolve active voxels.**
   For each active k: build $Q_k(\{\mu_\ell\}_{\ell\sim k})$ and advance $p_k \leftarrow e^{Q_k \Delta t} p_k$ (matrix exponential, Krylov method).
3. **Update equilibrated means.**
   $\mu_k^{\mathrm{loc}} \leftarrow (k_b + d \sum_{\ell\sim k} \mu_\ell)/(k_d + d|\mathrm{nbrs}|)$.
4. **Accumulate flux into empty voxels.**
   $\Phi_k \mathrel{+}= d\, \mu_\ell\, \Delta t$ for each active/equil. ℓ adjacent to empty k.
5. **Prolong (activate).**
   If $\Phi_k \geq \varepsilon_{\mathrm{exp}}$: set $p_k = \delta_0$, activate.
6. **Restrict (deactivate).**
   If $|\mu_k - \mu_k^{\mathrm{loc}}|/\mu_k^{\mathrm{loc}} < \varepsilon_{\mathrm{eq}}$ **and** $d_{\mathrm{TV}}(p_k, \mathrm{Pois}(\mu_k^{\mathrm{loc}})) < \varepsilon_{\mathrm{TV}}$: deactivate, store μ_k^{loc}.

Computational cost per step

Let K_{act} = active voxels, K_{eq} = equilibrated voxels.

Dominant cost: K_{act} independent CME solves, each of size $(n_{\max}+1) \times (n_{\max}+1)$.

At any time $K_{\mathrm{act}} \ll K$: only the **wavefront ring** is active simultaneously.

State-space comparison

Method	States tracked
Full FSP	$n_{\max}^{SK}$
Spatial adaptive FSP	$K_{\mathrm{act}} \cdot n_{\max}^{S}$

$K_{\mathrm{act}} \ll K$: exponential reduction.

2D birth-death: wavefront propagation and collapse

Model. $\emptyset \xrightarrow{k_b} A \xrightarrow{k_d} \emptyset$ in each voxel with diffusion D. Steady state: each voxel independent $\mathrm{Pois}(k_b/k_d)$.

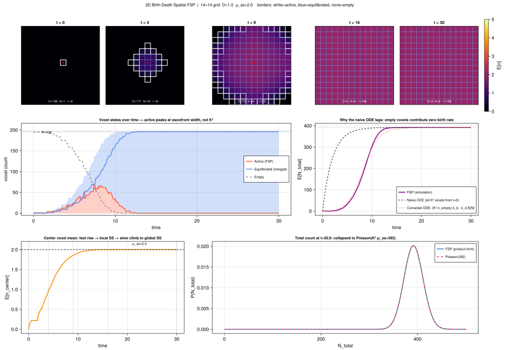
IC. All voxels empty; center voxel activated at $t = 0$. Birth reactions seed the wavefront.

Parameters: K=14, k_b=1, k_d=0.5, D=1, μ^*=2.

Voxel state counts over time:

- Active count peaks at $\sim$66 (one wavefront ring), **not** $K^2 = 196$.
- Equilibrated count grows monotonically behind the front.
- Active $\to$ 0: the distribution is *fully represented by coarse parameters*.

ODE comparison. Naive $K^2 k_b$ ODE *over*estimates: empty voxels do not birth. Corrected ODE



Center voxel: two-phase equilibration

The center voxel is activated at $t = 0$ and deactivated **early** once its distribution reaches its *local* SS.

Local SS (4 empty neighbours):

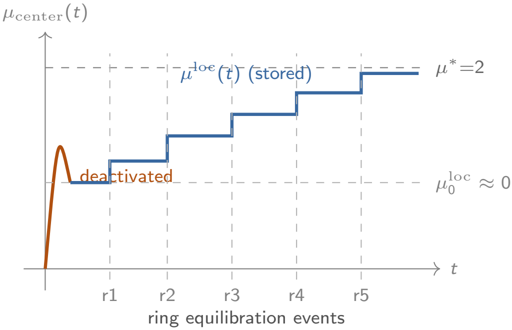
$$\mu^{\mathrm{loc}}_{\mathrm{center}} = \frac{k_b}{k_d + 4d} = \frac{1}{0.5 + 4} \approx 0.22.$$

The center's stored mean then **evolves upward** as ring-1, ring-2, . . . equilibrate behind the wavefront.

Two phases of the center mean:

1. **Fast:** $\mu \nearrow 0.22$ in $\sim 1/k_d^{\mathrm{eff}} \approx 0.2$ time units. Deactivated.
2. **Slow:** $\mu^{\mathrm{loc}} \nearrow 2.0$ over $\tau_{\mathrm{diff}} \approx (K/2)^2/(2D) \approx 24$ time units as successive rings equilibrate.

This **staircase** in μ^{loc} traces the equilibration wave



Collapse to a single well-mixed CME

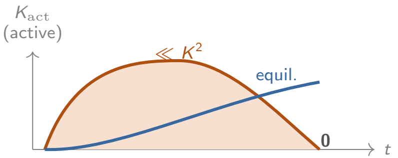
Once all voxels are equilibrated:

- All $\mu_k^{\mathrm{loc}} \to \mu^* = k_b/k_d = 2$.
- No active voxels remain; FSP support $= \emptyset$.
- Total count $N = \sum_k n_k$ is independent of spatial arrangement.

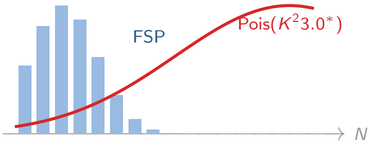
The **collapsed CME** for N is:

$$\emptyset \xrightarrow{K^2 k_b} N \xrightarrow{k_d} \emptyset,$$

with SS distribution $N^* \sim \mathrm{Pois}(K^2\mu^*)$.



t small: $p \approx \delta_N$ t large: $N \sim \mathrm{Pois}(K^2\mu^*)$



Product-form exactness

For birth-death, the voxels are independent at SS: $p^* = \prod_k \mathrm{Pois}(\mu^*)$. Total distribution is convolution of K^2 independent Poissons:

K^2

Summary and outlook

What we built:

- A spatially adaptive FSP for the RDME that maintains only a **sparse active set** — the current wavefront ring.
- Three-level representation: empty (δ_0), equilibrated ($\mathrm{Pois}(\mu^{\mathrm{loc}})$), active (full CME).
- Physics-driven prolong/restrict cycle: **an adaptive multigrid V-cycle in state space**.

Key properties:

- Computational work $\propto K_{\mathrm{act}} \cdot n_{\max}$, **not** $n_{\max}^K$.
- Mean-field coupling is exact at SS for linear kinetics.
- Equilibrated means evolve algebraically (local SS tracking).
- Total distribution collapses to $\mathrm{Pois}(K^2\mu^*)$.

Open directions:

1. **Nonlinear kinetics** (Schlögl, bistable networks).
   Mean-field is no longer exact. Need joint tracking at wavefront pairs (patch QSD).
2. **Joint wavefront tracking.**
   When K_{act} voxels are correlated, maintain a *joint* distribution rather than independent marginals.
3. **Adaptive re-activation.**
   Re-activate equilibrated voxels when perturbations arrive (e.g., front reflections, new injections).
4. **Higher dimensions and irregular grids.**
   The prolong/restrict logic generalises